

Chetankumar S Majjagi

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Competitive Programming Profiles: [Codeforces \(Max: 1601\)](#) • [CodeChef \(Max: 1829\)](#) • [LeetCode \(2228\)](#)

INTRODUCTION

Backend Developer diving deep into the oceans of distributed architectures and high-level system design. Passionate about building performance-focused, distributed, reliable, scalable applications and building & optimizing cloud Infra.

TECHNICAL SKILLS

- Languages: C/C++ (9.5/10), Python (8.5/10), JavaScript (8/10), SQL (9/10), Go (9/10)
- Web & Full Stack: React.js, Tailwind CSS, Python & FastAPI, PostgreSQL, MongoDB, Redis, WebSockets
- Tools, Infra & DevOps: AWS, GCP, Nginx, k6, Git, CI/CD Pipeline, Supabase, Pgbouncer, Docker Swarm, Terraform
- Core Concepts: Data Structures & Algorithms, CS Fundamentals, Concurrency & Multi-threading, System Design.

EDUCATION

International Institute of Information Technology, Naya Raipur

Bachelor of Technology in Computer Science Engineering (CGPA: 8.56/10.00)

Naya Raipur, India

2023 – 2027 (Expected)

PROJECTS

Tradeforces: Distributed HFT Benchmarking System [Git repo with detail Readme](#)

May 2026–Jun 2026

- Engineered an **8-microservice** evaluation platform, utilizing Kata Containers and AWS Firecracker to provision hardware-isolated **microVMs** in 200–300ms with strict CPU pinning.
- Automated runtime image compilation via **Google Cloud Build** and dynamic **K3s** pod generation, enforcing strict **pod affinity** to perfectly co-locate evaluation workloads.
- Developed a microsecond-precision **Go Telemetry Gateway** (7-goroutine pipeline) to dispatch raw **FIX protocol** orders over TCP. Used complex state-mapping data structures to track single-trade lifecycles and calculate execution latency.
- Architected a high-throughput event pipeline leveraging **Redpanda** for strict order serialization and **GCP Pub/Sub** for decentralized worker queues and dynamic backpressure management.
- **Tech Stack:** Go, FastAPI, React, GCS, Cloud Build, Redpanda, Pub/Sub, Postgres, Kubernetes (K3s), Docker, Terraform.

Ping: Distributed Real-Time Messaging System [Git repo with detail Readme](#)

Apr 2026–Present

- Engineered **Ping**, a real-time messaging system, by evolving a monolithic design into a scalable microservices architecture that decouples the main API from the connection manager.
- Developed a co-locating FastAPI and gRPC servers in a single Python application to share memory for WebSocket objects.
- Architected a **Distributed system**, enabling horizontal scaling with a Dynamic Nginx Load Balancer for the gateway and Redis for the replica mapping and finding active replicas.
- In a high-concurrency **k6 Load test** with backend deployed as a multi-replica in a multi-instance setup, the system handled **8000 concurrent virtual users** and **P95 latency of 198ms on 6000 vus** on a 5-core infra with Redis and Postgres GCP services.
- Tech Stack: Python, FastAPI, gRPC, Redis, ws, Postgres, k6, AWS, GCP, Nginx, CI/CD, PgBouncer, Docker Swarm, Terraform.

AI-Powered Relocation Survey Management System [Github](#)

Jan 2026–May 2026

- Digitalized the village relocation workflow, with the CV stage detection and a 4-step verification pipeline for transparency.
- Built an AI chatbot for **real-time dynamic data analysis** and visualization by plain-English queries, where the LLM generates pseudo-queries routed through a validator layer to enforce role-based authorization and enhanced security.
- Tech Stack: React, Flutter, Python, Flask, MongoDB, OpenCV, AI/LLM Integration.

ACHIEVEMENTS & CODING PROFILE

Competitive Programming

- Codeforces: Expert (1600+). 277, 956 ranks in contests.
- LeetCode: Guardian (2200+). Top 0.8% globally. Ranks: 311, 375, 569, 571 in contests.
- CodeChef: 4 star (1800+). Top 2% globally. Global ranks in contests: 137, 244.

Contest Performances

- ICPC 2025 Asia West Regionalist - Rank 100th/320 in Amritapuri region.
- Codefest IICPC Regionalist (2025): Ranked 203/17,000 in Round 1; 168/200 in Round 2.
- IIT Ropar COD-COM: Secured 11th place in the onsite final round.